“Pandas CSV Reader & Basic Analysis”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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❖ INTRODUCTION

The project titled “Pandas CSV Reader and Basic Analysis” focuses on understanding and analyzing structured data using the powerful Python library called Pandas. In the field of data science, handling datasets efficiently is essential and Pandas provides simple and effective tools for data manipulation and analysis.

In this project, a medical dataset containing information such as age, gender, department, diagnosis and treatment cost attributes is used for analysis. The primary aim is to read the dataset from a CSV file and convert it into a structured format known as a Data Frame. This makes it easier to access, modify and analyze the data.

The project demonstrates basic data analysis techniques such as inspecting the dataset using functions like head(), tail() and info() and understanding the data types of each column. It also involves calculating important statistical measures such as mean, median, minimum, maximum and count to summarize the dataset. These statistics help in identifying patterns and trends in medical treatment cost performance.

Additionally, the project includes filtering specific rows, selecting relevant columns and slicing subsets of the dataset to focus on meaningful insights. The processed data is then saved into a new CSV file for further use or reporting.

Overall, this project provides a basic understanding of how real-world medical data can be handled and analyzed using Pandas. It highlights the importance of data analysis in extracting useful information and supports better decision-making based on data.

❖ OBJECTIVES

1. To import the medical dataset and convert it into a Data Frame using Pandas.
2. To explore the structure of the dataset and understand its contents.
3. To perform basic statistical analysis to study medical dataset.
4. To extract useful information by applying filtering and selecting relevant columns.
5. To analyze specific sections of the dataset using slicing techniques.

❖ METHODOLOGY

The methodology of the project “Pandas CSV Reader and Basic Analysis” involves a systematic approach to handling and analyzing the medical dataset using the Pandas library in Python. The process is divided into several steps to ensure accurate and meaningful analysis of the data. Initially, the sales dataset is collected in CSV format. The data is then imported into the Python environment using the Pandas library and converted into a DataFrame, which provides a structured tabular format for easy manipulation and analysis.

In the next step, the dataset is inspected to understand its structure and contents. Functions such as head() and tail() are used to view the first and last few rows of the dataset, while dtypes and info() help identify the data types of each column and detect any missing or inconsistent values. This step ensures that the dataset is suitable for further analysis.

After inspection, basic statistical analysis is performed on the dataset. Key statistical measures such as mean, median, minimum, maximum and count are calculated to summarize the sales data and identify patterns or variations in sales and profit values. These statistics provide an overview of the distribution and trends within the dataset.

The next stage involves data filtering and selection. Specific rows are filtered based on certain conditions, such as higher sales values, to focus on relevant data. Additionally, important columns such as sales amount and profit are selected to simplify the analysis and highlight key variables. Slicing techniques are also applied to extract subsets of the dataset for detailed examination.

Finally, the processed and filtered data is saved into a new CSV file. This allows the results to be stored, shared and used for further analysis or reporting purposes.

Overall, this methodology ensures a structured and efficient approach to analyzing sales data and demonstrates how real-world datasets can be processed using Pandas to extract meaningful insights.

❖ Data Analysis

1. To import the sales dataset and convert it into a Data Frame using Pandas.

Code:

```
import pandas as pd

# Load Excel file

df = pd.read_ excel('Medical_Data.xlsx')

# Display first rows print(df.head())
```

Output:

	Age	Gender	Department	Diagnosis	Treatment Cost
0	43	Other	General Medicine	Diabetes	22964
1	2	Other	General Medicine	Hypertension	25233
2	24	Female	Neurology	Fracture	10575
3	65	Other	Cardiology	Migraine	11844
4	68	Other	Neurology	Fracture	34886

Interpretation:

The dataset is loaded using Pandas and converted into a DataFrame. The head() function displays the first five rows of the dataset.

2. To explore the structure of the dataset and understand its contents.

Code:

```
import pandas as pd

# Load dataset

df =

pd.read_excel(‘'Medical_Data.xlsx")

# Display first 5 rows print(df.head())

# Display last 5 rows

print(df.tail())

# Display data types

print(df.dtypes)
```

Output:

First Five Rows:

	Age	Gender	Department	Diagnosis	Treatment Cost
0	43	Other	General Medicine	Diabetes	22964
1	2	Other	General Medicine	Hypertension	25233
2	24	Female	Neurology	Fracture	10575
3	65	Other	Cardiology	Migraine	11844
4	68	Other	Neurology	Fracture	34886

Last Five Rows:

	Age	Gender	Department	Diagnosis	Treatment Cost
995	68	Male	Cardiology	Migraine	24392
996	14	Other	Neurology	Hypertension	6892
997	22	Male	Cardiology	Hypertension	47004
998	41	Male	Cardiology	Hypertension	22594
999	99	Other	Neurology	Diabetes	42653

Data Types:

Age	int 64
Gender	object
Department	object
Diagnosis	object
Treatment Cost	int 64

Interpretation:

The dataset contains both numerical and categorical data. The columns ‘Age’ and ‘Treatment Cost’ represent numerical values, while ‘Gender”, “Diagnosis” and ‘Diagnosis” are categorical. This structure makes the dataset suitable for basic data analysis.

3. To perform basic statistical analysis to study medical dataset.

Code:

```
# Summary statistics

print(df.describe())

# Mean values

mean_values = df.mean(numeric_only=True)

print("Mean:\n", mean_values)

# Median values

median_values = df.median(numeric_only=True)

print("Median:\n", median_values)
```

Output:

Statistic	Age	Treatment Cost
Count	1000	1000
Mean	49.87	25780.414
Median	50	25503.50
Minimum	1	1010
Maximum	99	49976

Interpretation:

The average Age is 49.87 and the average Treatment Cost is 25780.414. The minimum and maximum values show that Treatment Cost range from 1010 to 49976, indicating variation in cost. These statistics help in understanding overall Treatment Cost performance.

4. To extract useful information by applying filtering and selecting relevant columns.

Code:

```
# df[['Diagnosis','TreatmentCost']].max()

#df[['Diagnosis','Treatment Cost']].min()

#df[['Department','Treatment Cost']].max()

#df[['Department','Treatment Cost']].min()
```

Output:

Filtered Data1:

Diagnosis	Migraine
Treatment Cost	49976
dtype	object

Filtered Data 2:

Diagnosis	Diabetes
Treatment Cost	1010
dtype	object

Filtered Data3:

Diagnosis	Orthopedics
Treatment Cost	49976
dtype	object

Filtered Data4:

Diagnosis	Cardiology
Treatment Cost	1010
dtype	object

Interpretation:

The filtered data1 shows Migraine Diagnosis has the highest Treatment Cost. The filtered data2 shows Diabetes Diagnosis has the lowest Treatment Cost. The filtered data3 shows Orthopedics Department has the highest Treatment Cost. The filtered data4 shows Cardiology Department has the lowest Treatment Cost.

5. To analyze specific sections of the dataset using slicing techniques.

Code:

```
# Slice first 10 rows

subset1 = df.iloc[0:10]

# Slice specific rows and columns subset2 =

df.iloc[0:5][[ 'Department','Treatment Cost']]

# Display outputs
```

```
print("Subset 1 (First 10 Rows):")
print(subset1)
print("\nSubset 2 (Selected Rows & Columns):")
print(subset2)
```

Output:

Subset 1: First 10 Rows

	Age	Gender	Department	Diagnosis	Treatment Cost
0	43	Other	General Medicine	Diabetes	22964
1	2	Other	General Medicine	Hypertension	25233
2	24	Female	Neurology	Fracture	10575
3	65	Other	Cardiology	Migraine	11844
4	68	Other	Neurology	Fracture	34886
5	86	Male	General Medicine	Migraine	4391
6	56	Male	General Medicine	Hypertension	12355
7	6	Male	Cardiology	Migraine	3671
8	24	Other	Cardiology	Fracture	40818
9	11	Female	Cardiology	Migraine	47370

Subset 2: Selected Rows & Columns

Department	Treatment Cost
General Medicine	22964
General Medicine	25233
Neurology	10575
Cardiology	11844
Neurology	34886

Interpretation:

Slicing is used to extract specific rows and columns from the dataset. The first subset shows the initial records, while the second subset focuses only on important numerical columns for better analysis.

❖ Overall Conclusion

The project “Pandas CSV Reader and Basic Analysis” successfully demonstrates the fundamental concepts of data handling and analysis using the Pandas library in Python. Through this project, the medical dataset was effectively imported, organized, and analyzed in a structured Data Frame format, making it easier to work with and interpret the data. The use of functions such as head(), tail() and info() helped in understanding the structure and quality of the dataset, while statistical measures like mean, median, minimum, maximum and count provided meaningful insights into sales and profit trends.

Data filtering, column selection and slicing techniques were applied to extract relevant and useful information from the dataset, making the analysis more focused and clear. These operations helped in identifying patterns such as variations in sales values and differences in profit across different records.

Additionally, the project highlights the importance of basic data analysis techniques in real-world data science tasks. It shows how raw sales data can be transformed into meaningful information that supports better understanding and decision-making. In conclusion, this project builds a strong foundation in data analysis using Pandas and demonstrates how efficient tools can be used to extract valuable insights from business data.